

5. Use your scattergraph to comment on the *association* (relationship) between a person's height and the length of their arm span.
6. Draw a *trend* line through your graph. This is a straight line through the data points that approximately distributes the points equally above and below the line and reflects the relationship between the variables.
7. Use your trend line to predict the arm span of a person of height 172cm.
8. Comment on the original question posed: *can you predict the length of a person's arm span from their height?*

Bivariate Data

Often in a statistical investigation, more than one variable is needed to be collected. For example, in large health studies of populations it is common to obtain variables such as age, sex, height, weight, blood pressure, and total cholesterol about each individual. Economic studies may be interested in, among other things, personal income and years of education. Data for two variables is called bivariate data.

Association

The association between two variables is the relationship that can be described between them. The association can be classified by:

1. **form** – whether it is linear or non-linear
2. **direction** – whether it is positive or negative
3. **strength** – whether it is strong, moderate or weak

Trend line

A straight line can be fitted to a scattergraph that indicates the direction of the relationship. The line aims to go through the middle of the points such that there are an equal number of points above and below the line, and the points are as close to the line as possible. The trend line is used to make predictions.

Using the trend line to predict the value of a variable from within the given range of values, the process is called: **interpolation**.

Using the trend line to predict the value of a variable outside the given range of values, the process is called: **extrapolation**

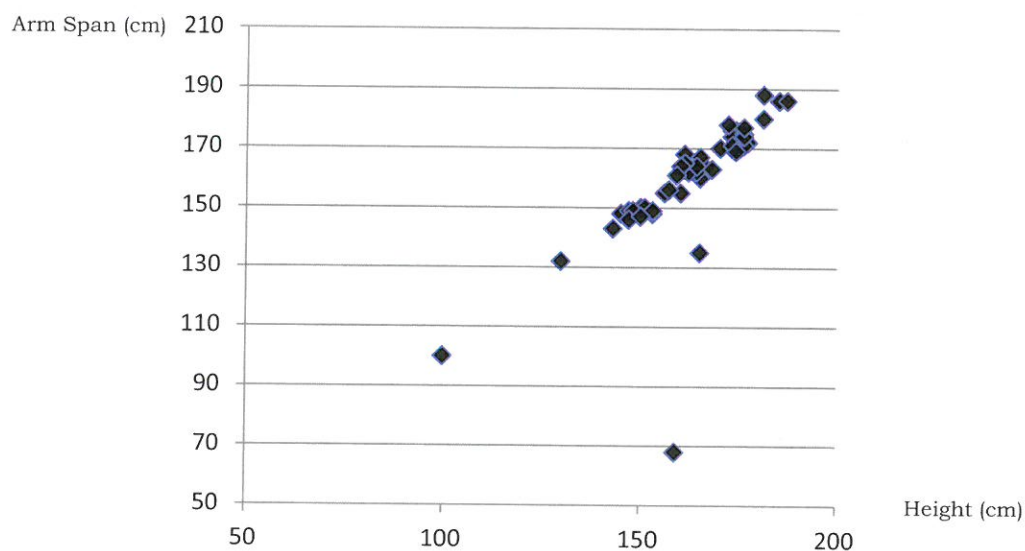
Factors affecting reliability of predictions

- The stronger the association the more accurate the prediction.
- Interpolation is generally more accurate than extrapolation.
- The number of observations (that is the number of points in the scatter plot) – generally the more observations (data points), the more reliable the prediction.

It is important to remember that association does not necessarily mean causation. Just because two variables show a relationship to each other, this does not mean that one causes the other, eg as the number of peas eaten by a person in his/her lifetime increases, so does the total number of hours they have slept. This does not mean that peas make you sleepy, nor that sleeping gives you a craving for peas. In this case, both are linked by a common factor of age.

Exercises

1. This scattergraph below shows the relationship between the arm span and height of 50 students.



- a) Using the three classifications of association, describe the relationship between the two variables.
 - b) One of the data points seem unlikely, locate this point and comment on factors that could have caused this result.
2. The table below shows the hours spent doing homework and playing video games per week of 15 Year 12 students.

Hours of homework	10	1	6	2	4	8	5	8	2	5	7	10	15
Hours on video games	0	9	4	6	13	3	7	1	10	1	1	6	3

- a) Choose a suitable scale and plot the data in a scatter graph.
 - b) Comment on the association between the variables.
 - c) Plot a trend line through your points.
3. The hours of paid work and the amount of income in a week for 18 students is displayed in the table.

Work	8	7	12	8	4	15	14	13	10
Income	100	150	100	60	45	200	200	120	165

Work	20	9	4	10	16	15	14	11	5
Income	389	150	44	184	190	180	130	120	90

- a) Use the data to draw a scatter graph of this information.
 - b) Comment on the association between the variables.
 - c) Draw a trend line through your graph.
 - d) Use your trend line to predict the income of a student who does 7 hours of paid work per week.
 - e) In full time employment, a person works approximately 38 hours per week. Use your trend line to predict the weekly pay of a person working full time.
 - f) Dave earned \$130 from his part time job. Use your trend line to predict the hours he worked.
 - g) Comment on the reliability of each of your predictions.
 - h) Explain why the data collected from this survey could have contained bias.
4. Using the three categories of association describe the relationship between the variables in each of the scatter graphs.

